Self-speculative decoding Stern et al. (2018) are, to the best of our knowledge, the first to suggest a speculative decoding scheme for faster inference. Our architecture replaces their linear prediction heads by transformer layers, but is otherwise similar. By reorganizing the order of the forward/backward, we can use all loss terms instead of stochastically picking one head for loss computation. Cai et al. (2024) present a more elaborate self-speculative decoding scheme that uses the top-k predictions of each head instead of the best one only. It can be used with the multi-token prediction models we train.

Multi-target prediction Multi-task learning is the paradigm of training neural networks jointly on several tasks to improve performance on the tasks of interest (Caruana, 1997). Learning with such auxiliary tasks allows models to exploit dependencies between target variables and can even be preferable in the case of independent targets (Waegeman et al., 2019). While more specifically tailored architectures for multi-target prediction are conceivable (Spyromitros-Xioufis et al., 2016; Read et al., 2021), modern deep learning approaches usually rely on large shared model trunks with separate prediction heads for the respective tasks (Caruana, 1997; Silver et al., 2016; Lample et al., 2022) like we do. Multi-target prediction has been shown to be a successful strategy in various domains, e.g. for learning time series prediction with more distant time steps in the future as auxiliary targets (Vapnik and Vashist, 2009) or for learning from videos with several future frames (Mathieu et al., 2016; Srivastava et al., 2016) or representations of future frames (Vondrick et al., 2016) as auxiliary targets.

7. Conclusion

We have proposed multi-token prediction as an improvement over next-token prediction in training language models for generative or reasoning tasks. Our experiments (up to 7B parameters and 1T tokens) show that this is increasingly useful for larger models and in particular show strong improvements for code tasks. We posit that our method reduces distribution mismatch between teacher-forced training and autoregressive generation. When used with speculative decoding, exact inference gets 3 times faster.

In future work we would like to better understand how to automatically choose n in multi-token prediction losses. One possibility to do so is to use loss scales and loss balancing (Défossez et al., 2022). Also, optimal vocabulary sizes for multi-token prediction are likely different from those for next-token prediction, and tuning them could lead to better results, as well as improved trade-offs between compressed sequence length and compute-per-byte expenses. Finally, we would like to develop improved auxiliary prediction losses that operate in embedding spaces (LeCun, 2022).

Impact statement

The goal of this paper is to make language models more compute and data efficient. While this may in principle reduce the ecological impact of training LLMs, we shall be careful about *rebound effects*. All societal advantages, as well as risks, of LLMs should be considered while using this work.

Environmental impact

In aggregate, training all models reported in the paper required around 500K GPU hours of computation on hardware of type A100-80GB and H100. Estimated total emissions were around 50 tCO2eq, 100% of which were offset by Meta's sustainability program.

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References

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, et al. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*, 2021.

Gregor Bachmann and Vaishnavh Nagarajan. The pitfalls of next-token prediction, 2024.

Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks, 2015.

Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. Piqa: Reasoning about physical commonsense in natural language, 2019.

Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple llm inference acceleration framework with multiple decoding heads, 2024.

Rich Caruana. Multitask learning. *Machine learning*, 28: 41–75, 1997.

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde, Jared Kaplan, Harri Edwards, Yura Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.

Nakhun Chumpolsathien. Using knowledge distillation from keyword extraction to improve the informativeness of neural cross-lingual summarization. Master's thesis, Beijing Institute of Technology, 2020.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Li Dong, Nan Yang, Wenhui Wang, Furu Wei, Xiaodong Liu, Yu Wang, Jianfeng Gao, Ming Zhou, and Hsiao-Wuen Hon. Unified language model pre-training for natural language understanding and generation. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*, pages 13063–13075, 2019.

Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi. High fidelity neural audio compression. *arXiv preprint arXiv:2210.13438*, 2022.

Moussa Kamal Eddine, Antoine J. P. Tixier, and Michalis Vazirgiannis. Barthez: a skilled pretrained french sequence-to-sequence model, 2021.

Alexander R. Fabbri, Irene Li, Tianwei She, Suyi Li, and Dragomir R. Radev. Multi-news: a large-scale multi-document summarization dataset and abstractive hierarchical model, 2019.

Mehrdad Farahani. Summarization using bert2bert model on wikisummary dataset. https://github.com/m3hrdadfi/wiki-summary, 2020.

Mehrdad Farahani, Mohammad Gharachorloo, and Mohammad Manthouri. Leveraging parsbert and pretrained mt5 for persian abstractive text summarization. In *2021 26th International Computer Conference, Computer Society of Iran (CSICC)*. IEEE, March 2021. doi: 10.1109/csicc52343.2021.9420563. URL http://dx.doi.org/10.1109/CSICC52343.2021.9420563.

Michael C Frank. Bridging the data gap between children and large language models. *Trends in Cognitive Sciences*, 2023.

Bogdan Gliwa, Iwona Mochol, Maciej Biesek, and Aleksander Wawer. Samsum corpus: A human-annotated dialogue dataset for abstractive summarization. In *Proceedings of the 2nd Workshop on New Frontiers in Summarization*. Association for Computational Linguistics, 2019. doi: 10.18653/v1/d19-5409. URL http://dx.doi.org/10.18653/v1/D19-5409.

Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh Nagarajan. Think before you speak: Training language models with pause tokens, 2023.

Dan Hendrycks, Steven Basart, Saurav Kadavath, Mantas Mazeika, Akul Arora, Ethan Guo, Collin Burns, Samir Puranik, Horace He, Dawn Song, et al. Measuring coding challenge competence with apps. *arXiv preprint arXiv:2105.09938*, 2021.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration, 2020.

Jianyu Zhang Leon Bottou. Multi-label classification as an auxiliary loss for language modelling. personal communication, 2024.

Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension, 2017.

Diederik Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *ICLR*, 2015.

Ryan Koo, Minhwa Lee, Vipul Raheja, Jong Inn Park, Zae Myung Kim, and Dongyeop Kang. Benchmarking cognitive biases in large language models as evaluators. *arXiv preprint arXiv:2309.17012*, 2023.

Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Matthew Kelcey, Jacob Devlin, Kenton Lee, Kristina N. Toutanova, Llion Jones, Ming-Wei Chang, Andrew Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. Natural questions: a benchmark for question answering research. *Transactions of the Association of Computational Linguistics*, 2019.

Guillaume Lample, Marie-Anne Lachaux, Thibaut Lavril, Xavier Martinet, Amaury Hayat, Gabriel Ebner, Aurélien Rodriguez, and Timothée Lacroix. Hypertree proof search for neural theorem proving, 2022.

Yann LeCun. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. *Open Review*, 62(1), 2022.

Benjamin Lefaudeux, Francisco Massa, Diana Liskovich, Wenhan Xiong, Vittorio Caggiano, Sean Naren, Min Xu, Jieru Hu, Marta Tintore, Susan Zhang, Patrick Labatut, and Daniel Haziza. xformers: A modular and hackable transformer modelling library. https://github.com/facebookresearch/xformers, 2022.

Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding, 2023.

Yujia Li, David Choi, Junyoung Chung, Nate Kushman, Julian Schrittwieser, Rémi Leblond, Tom Eccles, James Keeling, Felix Gimeno, Agustin Dal Lago, et al. Competition-level code generation with alphacode. *Science*, 378(6624):1092–1097, 2022.

Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, July 2004. Association for Computational Linguistics. URL https://aclanthology.org/W04-1013.

Zhenghao Lin, Zhibin Gou, Yeyun Gong, Xiao Liu, Yelong Shen, Ruochen Xu, Chen Lin, Yujiu Yang, Jian Jiao, Nan Duan, and Weizhu Chen. Rho-1: Not all tokens are what you need, 2024.

Ilya Loshchilov and Frank Hutter. Sgdr: Stochastic gradient descent with warm restarts, 2017.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization, 2019.

Michael Mathieu, Camille Couprie, and Yann LeCun. Deep multi-scale video prediction beyond mean square error, 2016.

Ramesh Nallapati, Bowen Zhou, Cicero Nogueira dos santos, Caglar Gulcehre, and Bing Xiang. Abstractive text summarization using sequence-to-sequence rnns and beyond, 2016.

Shashi Narayan, Shay B. Cohen, and Mirella Lapata. Don't give me the details, just the summary! topic-aware convolutional neural networks for extreme summarization, 2018.

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html.

OpenAI. Gpt-4 technical report, 2023.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback, 2022.

Koyena Pal, Jiuding Sun, Andrew Yuan, Byron C. Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state, 2023.

Weizhen Qi, Yu Yan, Yeyun Gong, Dayiheng Liu, Nan Duan, Jiusheng Chen, Ruofei Zhang, and Ming Zhou. Prophetnet: Predicting future n-gram for sequence-to-sequence pre-training, 2020.

Jesse Read, Bernhard Pfahringer, Geoffrey Holmes, and Eibe Frank. Classifier chains: A review and perspectives. *Journal of Artificial Intelligence Research*, 70:683–718, 2021.

Melissa Roemmele, Cosmin Adrian Bejan, and Andrew S Gordon. Choice of plausible alternatives: An evaluation of commonsense causal reasoning. In *2011 AAAI Spring Symposium Series*, 2011.

Maarten Sap, Hannah Rashkin, Derek Chen, Ronan LeBras, and Yejin Choi. Socialiqa: Commonsense reasoning about social interactions, 2019.

David Silver, Aja Huang, Chris J Maddison, Arthur Guez, Laurent Sifre, George Van Den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, et al. Mastering the game of go with deep neural networks and tree search. *nature*, 529(7587): 484–489, 2016.

Aaditya K Singh, Stephanie CY Chan, Ted Moskovitz, Erin Grant, Andrew M Saxe, and Felix Hill. The transient nature of emergent in-context learning in transformers. *arXiv preprint arXiv:2311.08360*, 2023.

Eleftherios Spyromitros-Xioufis, Grigorios Tsoumakas, William Groves, and Ioannis Vlahavas. Multi-target regression via input space expansion: treating targets as inputs. *Machine Learning*, 104:55–98, 2016.

Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video representations using lstms, 2016.

Mitchell Stern, Noam Shazeer, and Jakob Uszkoreit. Blockwise parallel decoding for deep autoregressive models, 2018.

Yi Tay, Mostafa Dehghani, Vinh Q Tran, Xavier Garcia, Jason Wei, Xuezhi Wang, Hyung Won Chung, Siamak Shakeri, Dara Bahri, Tal Schuster, et al. Ul2: Unifying language learning paradigms. *arXiv preprint arXiv:2205.05131*, 2022.

Vladimir Vapnik and Akshay Vashist. A new learning paradigm: Learning using privileged information. *Neural networks*, 22(5-6):544–557, 2009.

Carl Vondrick, Hamed Pirsiavash, and Antonio Torralba. Anticipating visual representations from unlabeled video, 2016.

Willem Waegeman, Krzysztof Dembczyński, and Eyke Hüllermeier. Multi-target prediction: a unifying view on problems and methods. *Data Mining and Knowledge Discovery*, 33:293–324, 2019.

Vikas Yadav, Steven Bethard, and Mihai Surdeanu. Quick and (not so) dirty: Unsupervised selection of justification sentences for multi-hop question answering. *arXiv preprint arXiv:1911.07176*, 2019.

Zhilin Yang, Zihang Dai, Yiming Yang, Jaime Carbonell, Russ R Salakhutdinov, and Quoc V Le. Xlnet: Generalized autoregressive pretraining for language understanding. In *Advances in neural information processing systems*, pages 5753–5763, 2019.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. Hellaswag: Can a machine really finish your sentence?, 2019.